SPX Sort-to-Light

Procurement Specification

Prepared by: Local Dev Team
Purpose: Procurement — Hardware Specification
Version: v1.0

Table of Contents

Table of Contents 1
1. Architecture 2
2. Hardware Requirements 2
2.1 Overall 2
2.2 LED Strip 3
2.3 Electrical Line with Connector 5
2.4 Chip Controller 7
2.4.1 AC → DC Converter 7
2.4.2 PCB Controller 8
2.5 Quantity Requirement per Rack 8
2.6 Quantity Requirement for Pilot 8
3. Software Requirements 11

1. Architecture

This is the overall architecture of the Sort-to-Light project.

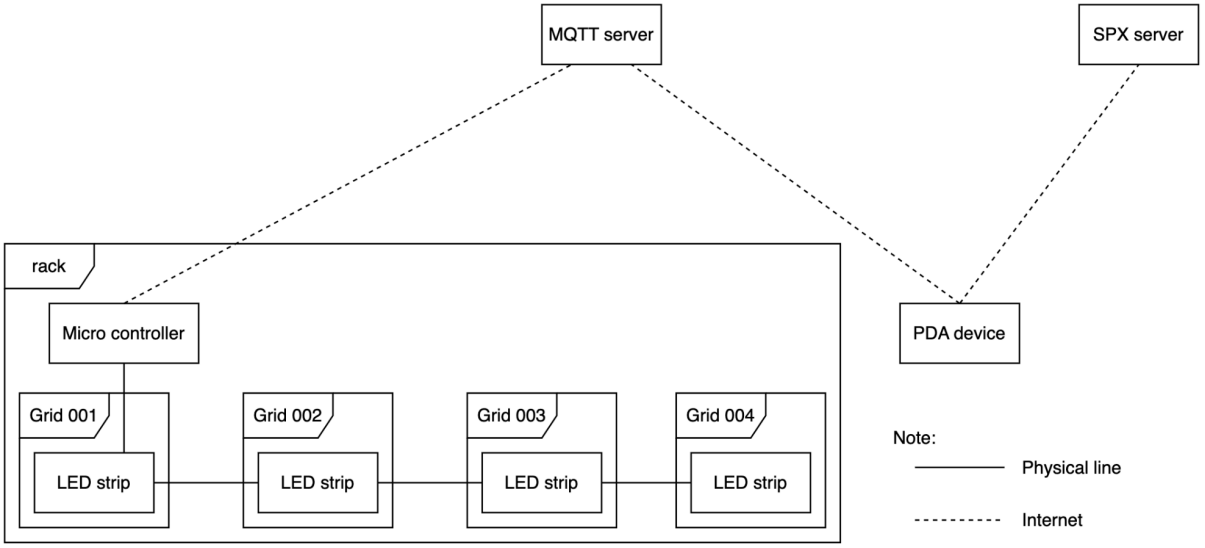


Figure 1: Overall Sort-to-Light System Architecture

The following diagram shows, step by step, the flow from scanning the waybill number to turning on the light.

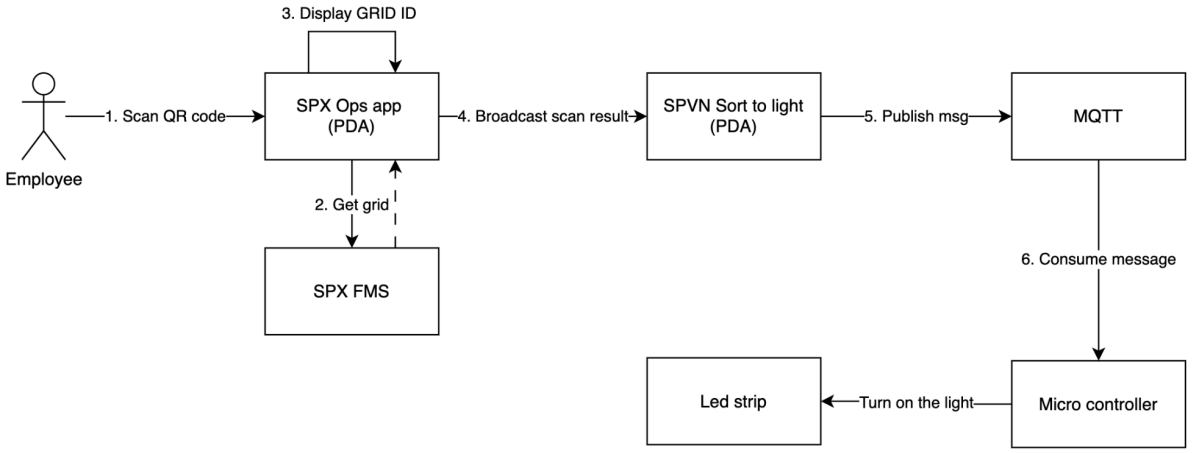


Figure 2: Scan-to-Light Process Flow

2. Hardware Requirements

2.1 Overall

This is an example of the rack.



Figure 3: Example Sorting Rack

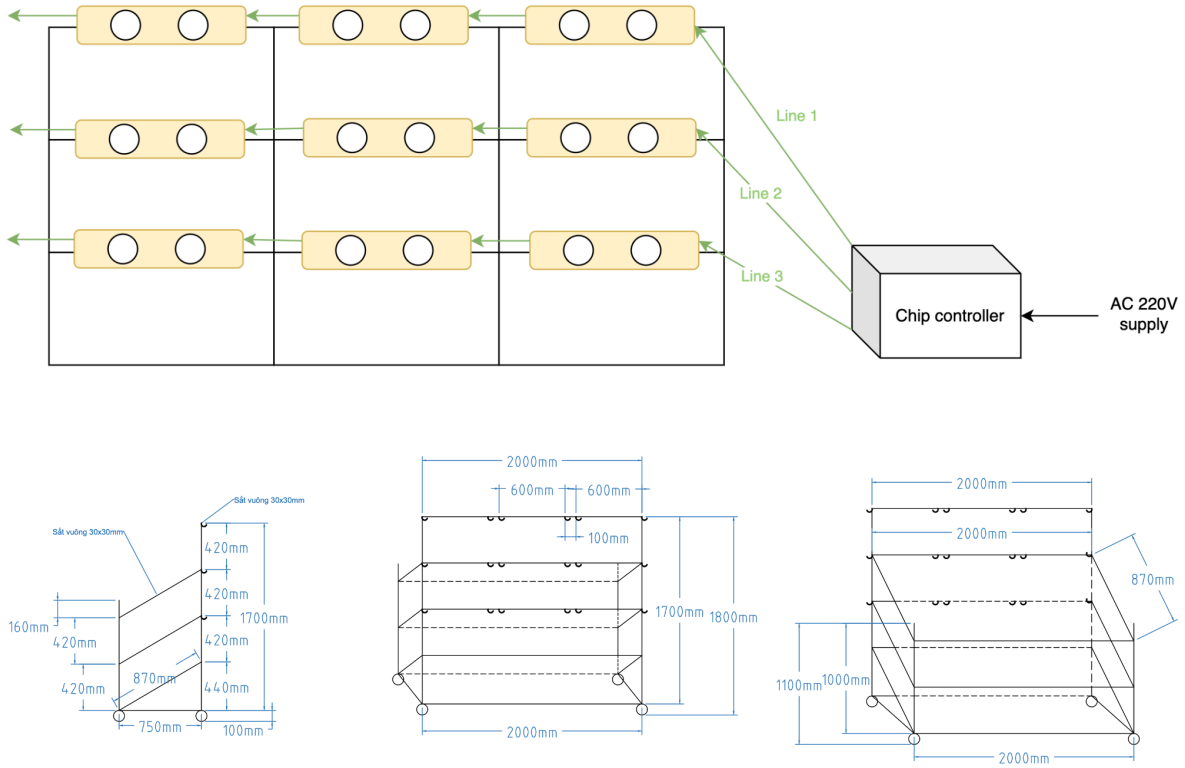


Figure 4: Rack Dimensions

This is how the LED strips will be arranged on the rack.

Figure 5: LED Arrangement per Rack

Each rack will have 1 chip controller and 3 LED strips. Each LED strip is a row on the rack. The chip controller will use an AC 220V power supply.

2.2 LED Strip

Each of the following images is called a LED module.

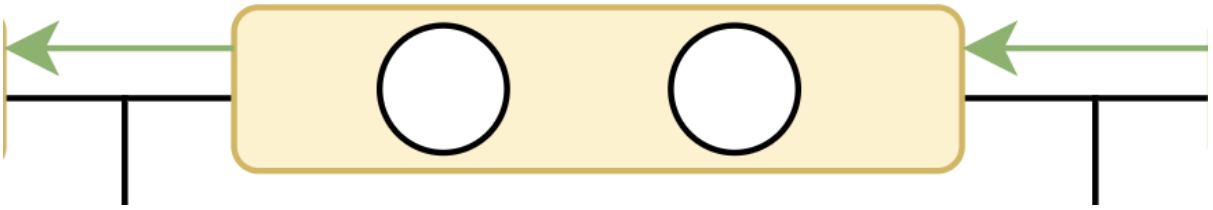


Figure 6: LED Module

A LED module consists of:

- LED hắc ba mắt full color IC16703 (https://ngocbaolong.vn/san-pham/led-hat-3-bong-full-color-16703/)
- Thanh nhôm LED định hình kèm nắp chụp mica



Figure 7: Thanh nhôm LED định hình kèm nắp chụp mica

- Bát khoá chữ nhật ôm trọn mặt mica
- Nắp đậy mica với đầu connector 5557, 5559 4.2mm 4 chân

Dimensions

This is the size of a LED:

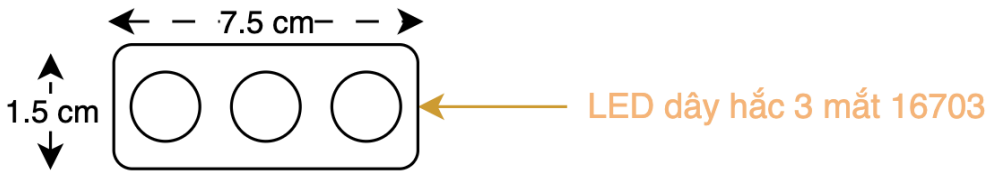


Figure 8: LED Dimensions

This is the size of a LED module:

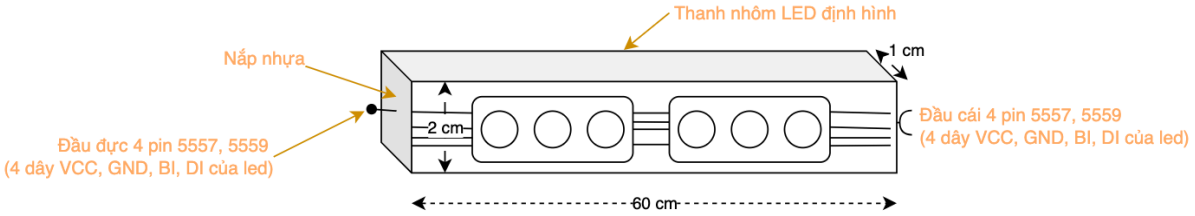


Figure 9: LED Module Dimensions

This is the size of “thanh nhôm LED định hình”:

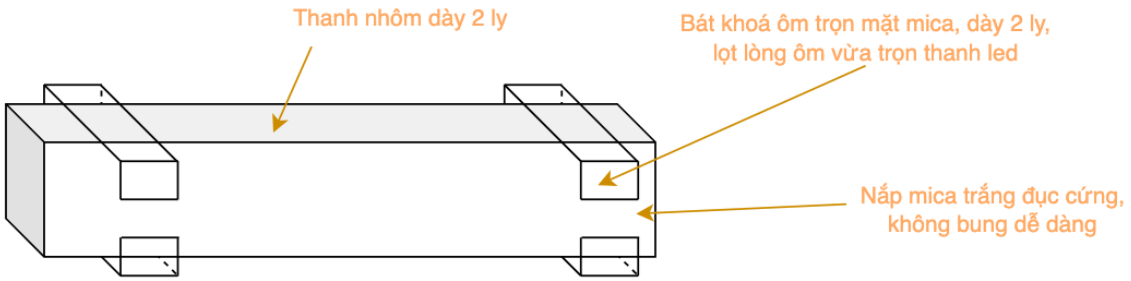


Figure 10: Thanh nhôm LED định hình — Dimensions

This is the size of “Bát khoá”:

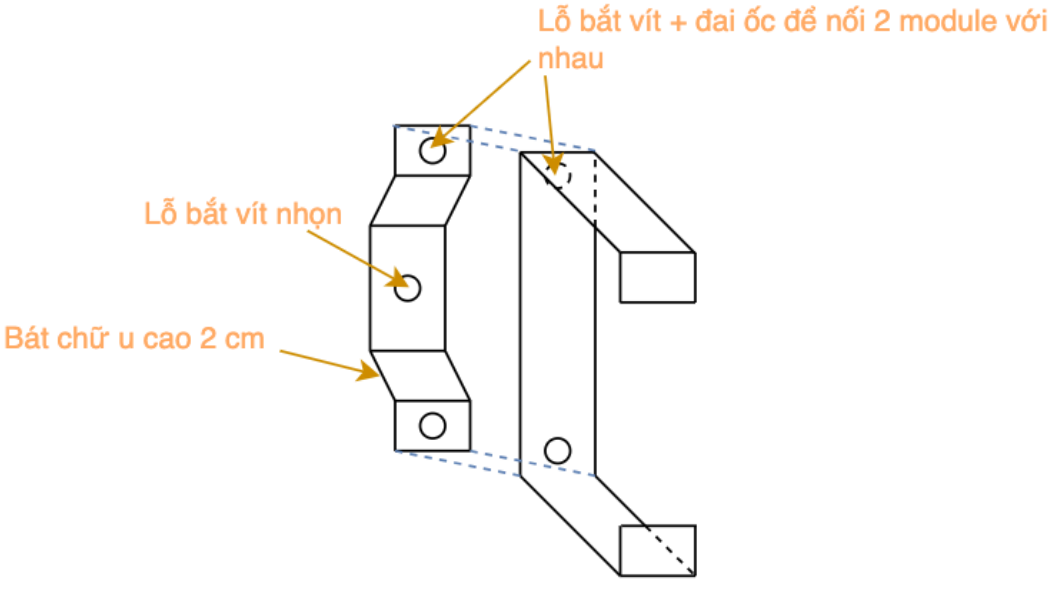


Figure 11: Bát khoá — Dimensions

2.3 Electrical Line with Connector

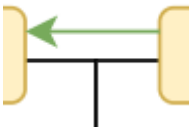


Figure 12: Connector Detail

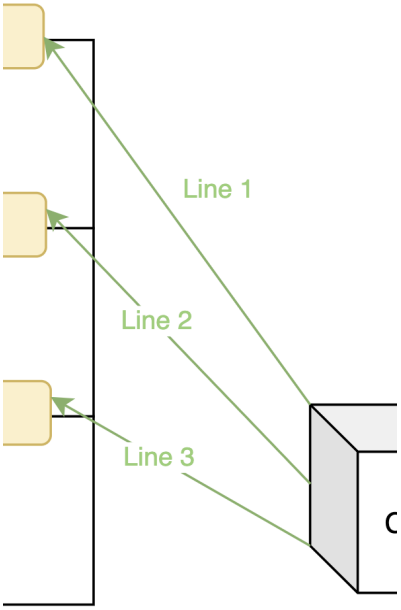


Figure 13: Electrical Line Assembly

There are 2 electrical lines:

- Type 1: both connectors of the line are 5557, 5559

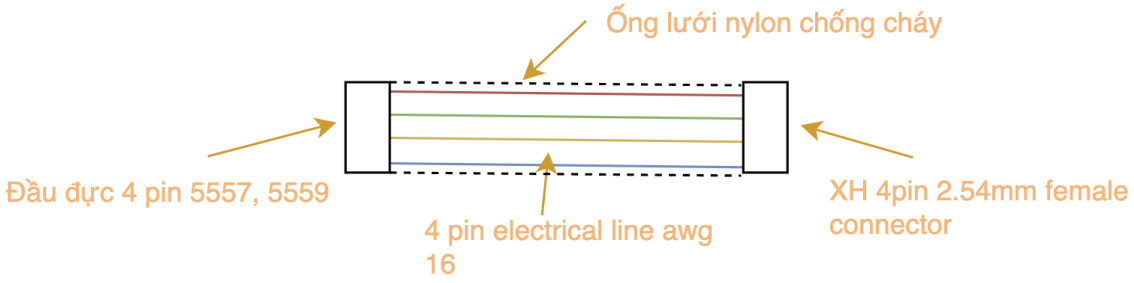


Figure 14: Type 1 Electrical Line

- Type 2: the female connector is VH 3.96mm 4 pin

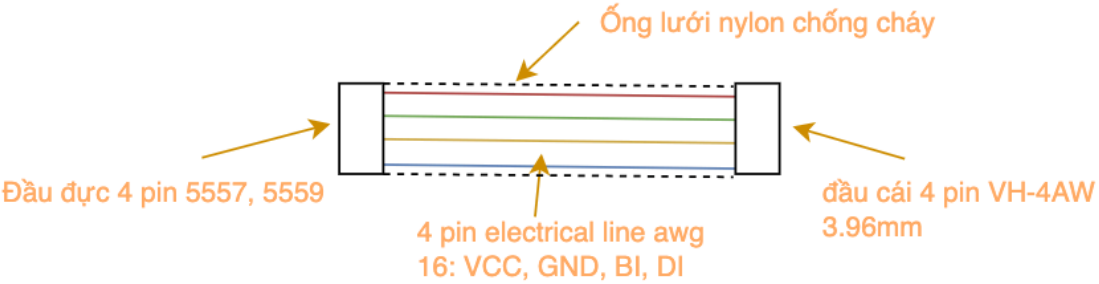


Figure 15: Type 2 Electrical Line

Note:

- The male VH connector is put in the board.

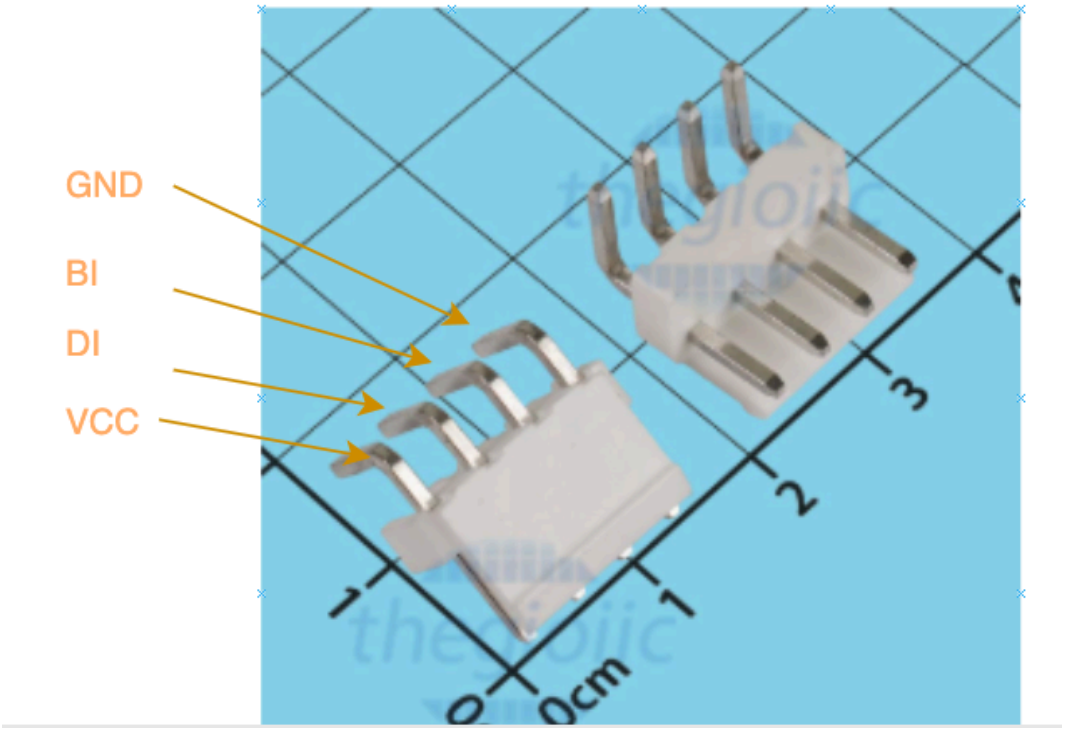


Figure 16: Male VH Connector Placement on Board

- The LED strip has data direction and pin (VCC, DI, BI, GND) orders.
- Other connectors must respect the order of the LED strip and board connector.

2.4 Chip Controller

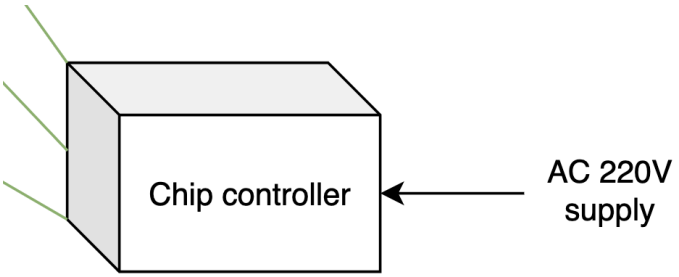
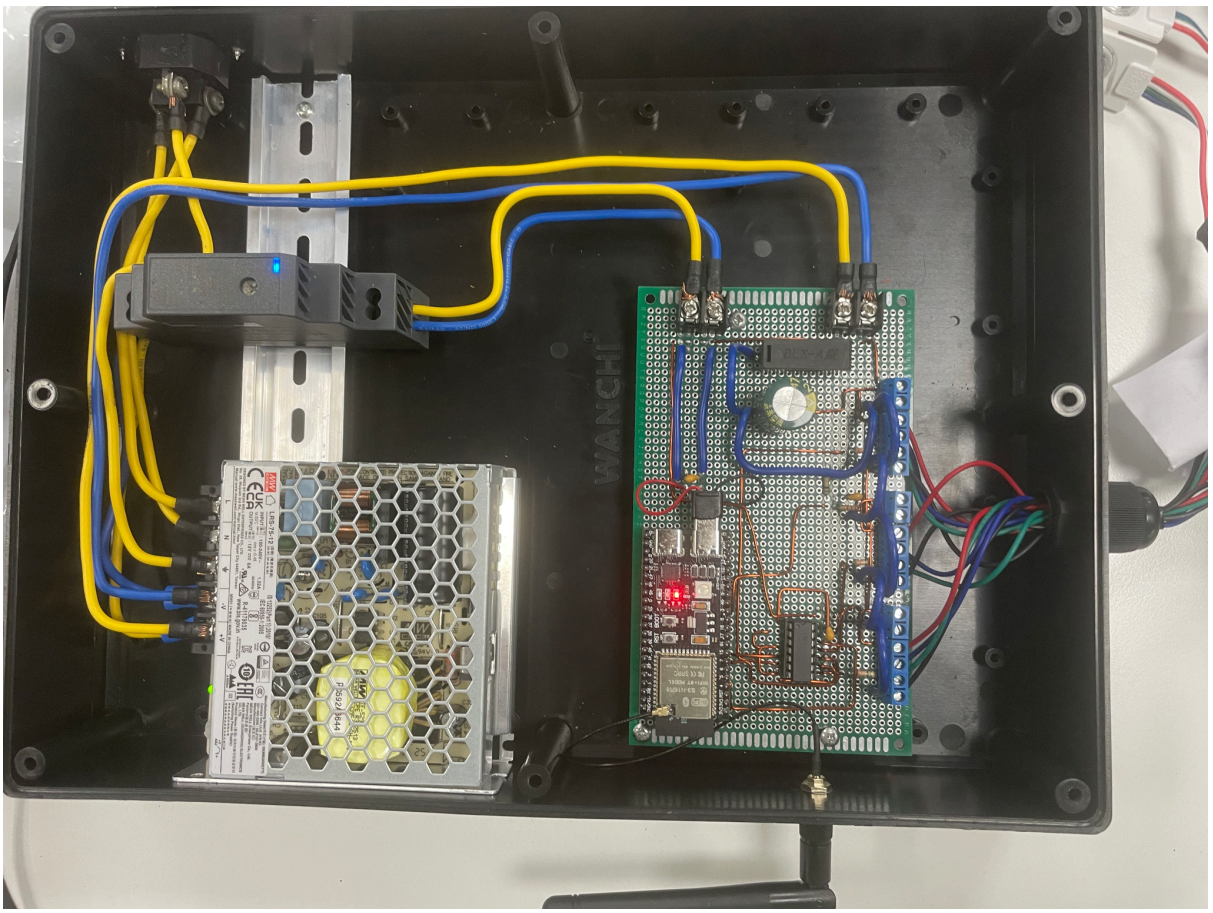


Figure 17: Chip Controller Assembly



2.4.1 AC → DC Converter

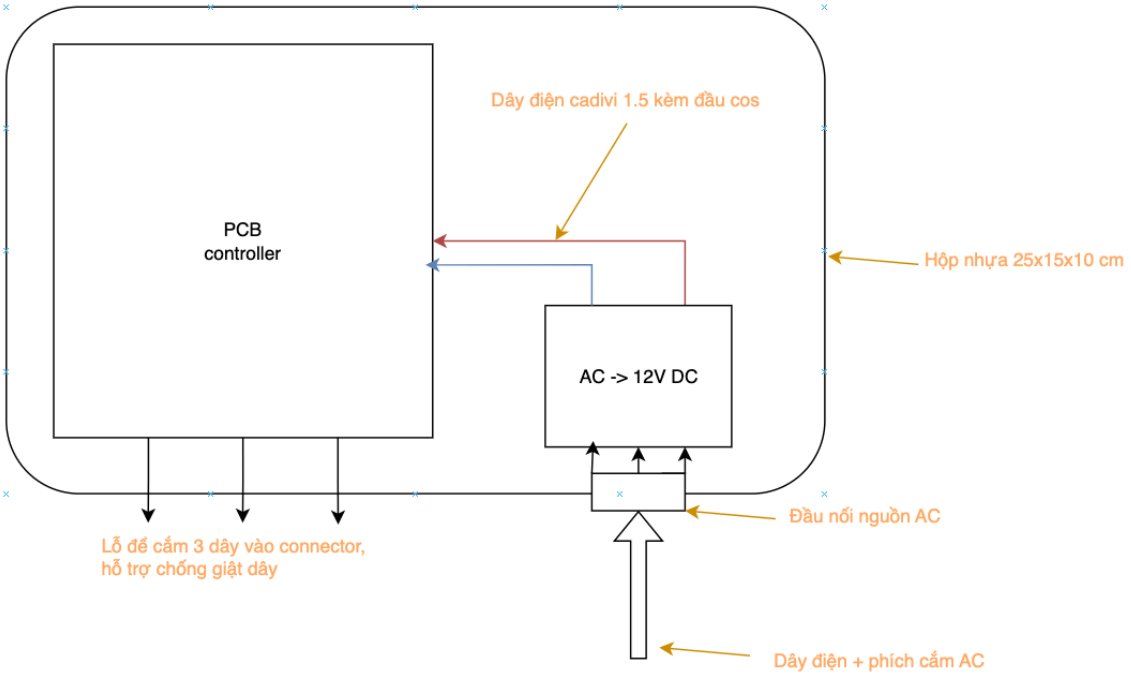


Figure 18: AC → DC Converter

2.4.2 PCB Controller

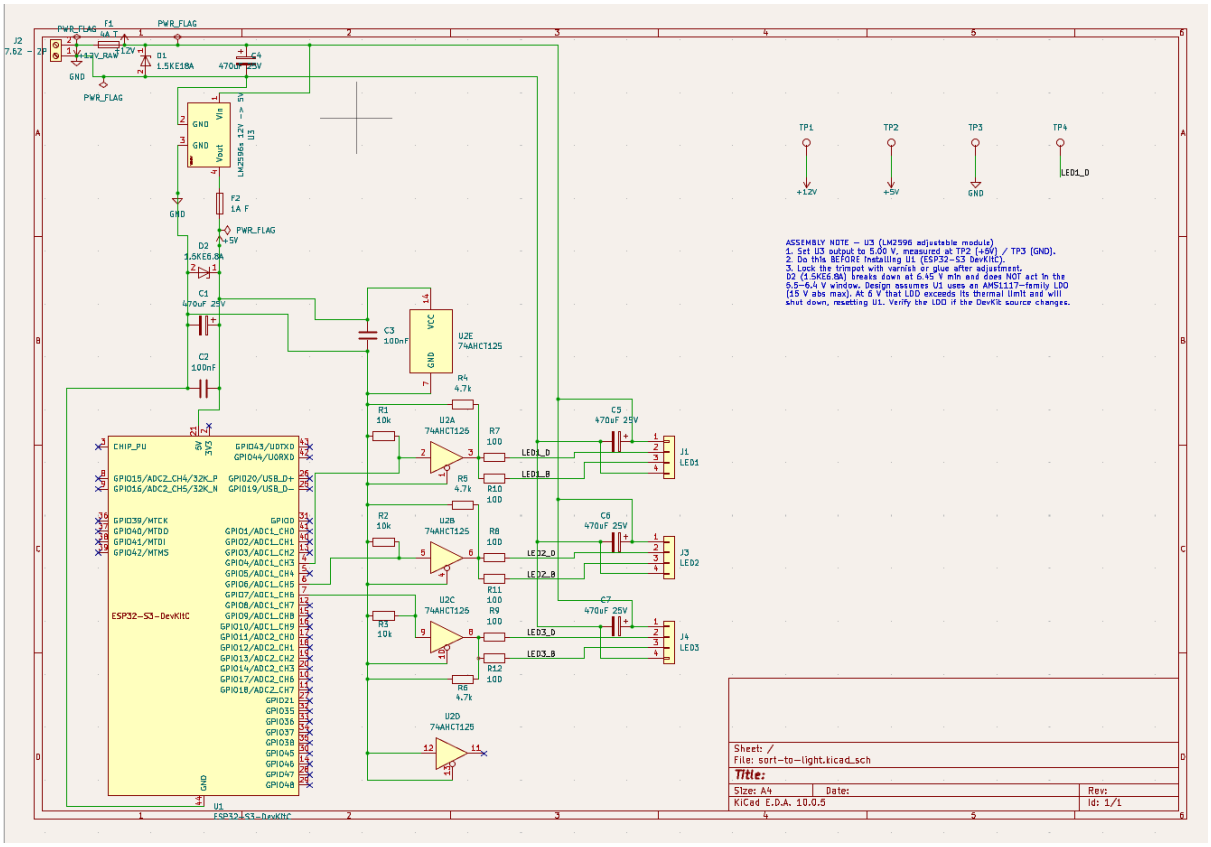


Figure 19: PCB Controller

Check the attached schematic design for more detail.

2.5 Quantity Requirement per Rack

Category	Name	Amount	Example	Example price
Led strip module	Led module *LED hắt 3 bóng RGB SMD 5050 (DC 12V)*	18		650.000
Electrical line	Type 1 15 cm	10		60.000
	Type 1 35 cm	10		80.000
	Type 2 2m	3		50.000
	Type 2 1m5	3		50.000
	Type 2 1m	3		50.000
Chip controller	Module (có đầy đủ PCB, hộp nhựa, connector, dây điện, etc)	1		
The list below is just for reference for implementing a PCB controller				
	Dây điện AC	1	https://www.thegioiic.com/day-cap-nguon-ac-10a-220v-dai-	80.000

Category	Name	Amount	Example	Example price
			1-5m-loi-3x0-75mm2-c13-iec-noi-cee nếu không có nối đất https://www.thegioiic.com/day-cap-nguon-ac-10a-250v-dai-1-5m-loi-dong-3x0-75mm2-c13-iec-noi-phich-cam-us-3-chan nếu có nối đất	
	Đầu nối nguồn AC	1	https://www.thegioiic.com/ac-04-dau-noi-nguon-ac-gan-bang-chan-dong-bat-vit-250v-10a	10.000
	Hộp nhựa có mounting plate	1	https://vi.made-in-china.com/co_kitocici/product_150X250X100-IP68-Plastic-Distribution-Box-Transparent-Lid-Waterproof-Junction-Box-Factory-Price_hroigyiyg.html	145.000
	Dây điện cadivi 1.5 màu vàng	2m	https://www.thegioiic.com/day-dien-don-cadivi-cv-1-5-1-5mm2-mau-vang	21.660
	Dây điện cadivi 1.5 màu xanh dương	2m	https://www.thegioiic.com/day-dien-don-cadivi-cv-1-5-1-5mm2-mau-xanh-duong	21.660
	Đầu cos chữ y	20	https://www.thegioiic.com/sv1-25-4-dau-cos-chu-y-co-vo-day-0-5-1-5mm2-vit-m4-mau-den	9.120
	Tru đồng đỡ PCB	4	https://www.thegioiic.com/tru-dong-hex-m3-duc-cai-dai-7mm	5.472
	Vít nhon M4 20mm	20	https://www.thegioiic.com/vit-nhon-ka-m4-x-10mm-304	18.240
	Vít đầu dù m3 6mm	20	https://www.thegioiic.com/vit-dau-du-tm-m3-x-6mm	6.620
	Nguồn AC - DC 12V Mean Well LRS-75-12	1	https://qaled.com.vn/san-pham/nguon-meanwell-lrs-75/nguon-meanwell-lrs-75-12-12vdc-6a.html	230.000
	Ốc siết cáp nhựa	3	https://www.thegioiic.com/pg11-oc-siet-cap-nhua-chong-nuoc-ip68-co-dai-oc-mau-trang	6.498
	PCB đục lỗ FR4	1	https://www.thegioiic.com/fr4-pcb-han-test-board-9x15cm-phu-xanh-2-mat	41.368
	KF7.62-2 Terminal Khối Hàn PCB 2 Cực 20A 300V 7.62mm	1	https://www.thegioiic.com/kf7-62-2-terminal-khoi-han-pcb-2-cuc-20a-300v-7-62mm	1.824
	Cầu chì 4A	5	https://www.thegioiic.com/cau-chi-su-4a-5x20mm	5.700

Category	Name	Amount	Example	Example price
	Đế cầu chì	2	https://www.thegioiic.com/blx-a-de-cau-chi-5x20mm-6a-250vac	5.130
	Tu hoá 470	5	https://www.thegioiic.com/tu-hoa-470uf-25v-8x12mm-xuyen-lo	6.270
	Tu gốm 104 100nF	2	https://www.thegioiic.com/tu-gom-multilayer-100nf-0-1uf-50v	4.560
	Trở 10k	3	https://www.thegioiic.com/dien-tro-10-kohm-1-4w-5-4-vong-mau	2.750
	Trở 4.7k	3	https://www.thegioiic.com/dien-tro-4-7-kohm-1-4w-5-4-vong-mau	2.750
	Trở 100	6	https://www.thegioiic.com/dien-tro-100-ohm-1-4w-5-4-vong-mau	2.750
	Hàng rào cái đơn 40 chân	2	https://www.thegioiic.com/hang-rao-cai-don-2-54mm-40-chan-1-hang-cao-11-8mm-xuyen-lo	5.244
	Đế cắm IC DIP14	1	https://www.thegioiic.com/de-ic-14-chan-vuong-hep-hai-hang-2-54mm-1a-60vac	8.550
	Chì hàn	1	https://www.thegioiic.com/thiec-han-0-5-sn63-pb37-55g-mechanic	104.244
	Băng keo điện	1	https://www.thegioiic.com/bang-keo-dien-nano-den-rong-18mm-dai-18m	7.980
	MCU controller ESP32-S3-WROOM-1, 16MB Flash, 8MB PSRAM loại có 2 angten	1	https://dientuduchuy.com/products/esp32-s3-n16r8-esp32-s3-devkitc-1 hoặc https://hshop.vn/mach-phat-trien-mke-k01-esp32-s3-dev-kit	295.000
	Angten 2.4GHz cho chip	1	https://dientuduchuy.com/products/angten-sma-2-4ghz	22.000
	Data level shifter 74AHCT125	1	https://shopee.vn/product/299128494/52161092311	50.078
	Dây điện 1.0 lõi đồng đơn daphaco	2m		14.000
	TVS 1.5KE18CA	4	https://www.thegioiic.com/1-5ke18ca	13.680
	P6KE6.8AG	4	https://www.thegioiic.com/p6ke6-8ag	6.840
	Đầu VH connector	3	https://www.thegioiic.com/dau-vh-4aw-3-96mm-4-chan-cong-xuyen-lo	4.446

Category	Name	Amount	Example	Example price
Total Estimation				2099.434 VND

2.6 Quantity Requirement for Pilot

SOC	Number of Racks in the Pilot	Number of Racks + Buffer (30%)
SOC BMT	3	4
SOC Vinh	3	4
SOC Dien Khanh	3	4
SOC HY Temp	5	7
SOC HN	5	7
Total	19	**26**

➢ Total Number of Racks in the Pilot: **26 racks** (including buffer)